

MMAE 443 SYLLABUS  
**SYSTEM ANALYSIS AND CONTROL**  
Fall 2025



 TBA

**COURSE TEXT:** Franklin, G. F., Powell, J. D., and Emami-Naeni, A., *Feedback Control of Dynamic Systems*, 8<sup>th</sup> Edition, Addison-Wesley, 2018. (Older editions are also acceptable.)

**PREREQUISITES:** MMAE 305 and MATH 252

- GOALS:** After completing this course, you should be able to:
- Develop mathematical models (linear or linearized differential equations) of dynamic systems
  - Apply the Laplace transform to generate transfer functions and to compute the time and frequency response for a dynamic system
  - Understand the meaning of stability and the nature and purpose of feedback (closed-loop) control
  - Apply the fundamental classical methods of control system design
  - Design feedback control systems to meet specified performance criteria

**GRADING POLICY:**

|                   |     |
|-------------------|-----|
| Homework (weekly) | 20% |
| Exam 1            | 25% |
| Exam 2            | 25% |
| Final Project     | 30% |

**HOMEWORK:** Weekly homework will be completed via a Blackboard quiz and will be due by midnight on the due date. The quizzes can be taken two times and only the final attempt will count toward your grade.  
**Late homework policy:** Homework assignments are due on the date listed. Ten points will be taken every day the assignment is late. Assignments over one week late will not be accepted.

**EXAMINATIONS:** There will be two exams during the semester. If you are ill (with acceptable medical proof from a physician), have an emergency (with proof), or have a conflict, you must resolve this issue with the instructor as soon as possible prior to the exam.

**FINAL PROJECT:** There will be a final project that requires you to use the analysis and control skills you learned to understand and design a control system for a real-life example. This project will be done as a group and presented orally in the final weeks of the semester.

**COMPUTER USAGE:** Students will be expected to use MATLAB for some of the homework assignments. You can find a Get Started Guide for MATLAB at:  
[http://www.mathworks.com/access/helpdesk/help/pdf\\_doc/matlab/getstart.pdf](http://www.mathworks.com/access/helpdesk/help/pdf_doc/matlab/getstart.pdf)

**POLICY ON ACADEMIC HONESTY:** The Department of Mechanical, Materials, and Aerospace Engineering at Illinois Institute of Technology considers academic training to be apprenticeship for practice in the professions. Students are expected to demonstrate a code of moral integrity and ethical standards commensurate with the high expectations that society places upon professional practice.

Accordingly, it is the policy of the department to maintain the highest standard of academic honesty and integrity. The department prohibits any form of cheating that contributes, or is intended to contribute, to a student's grade in a course. This includes copying or unauthorized collaboration on homework, reports, computer programs, projects, etc., collaboration of any kind on an examination or quiz, or the unauthorized possession or use of any material or machine-stored information during an examination. It is a violation for a student, whether or not currently enrolled in the university, to knowingly engage or attempt to engage in the acquisition, without permission, of tests, answer sheets, problem solutions or other academic material when such material has been withheld from distribution by the instructor.

Students judged guilty of such offenses are subject to either of the following sanctions:

1. **Expulsion from a course.** The student is assigned a punitive failing grade of E for the course and can no longer participate in the course or receive evaluation of coursework from the instructor.
2. **Reduction in Grade.** A reduction in grade for the assignment or exam involved or for the course may be applied.

In addition, the incident will be reported to the Designated Dean for Academic Discipline, who may make a recommendation for further disciplinary action. The report of the incident will be placed in the student's permanent academic record. An offense may warrant a recommendation to initiate dismissal proceedings.

This policy will apply not only to a student guilty of any form of academic dishonesty, but also to any party who knowingly allows his or her work to be plagiarized.

This policy is consistent with the IIT Code of Academic Honesty  
([http://www.iit.edu/student\\_affairs/handbook/](http://www.iit.edu/student_affairs/handbook/))

## MMAE 443 TENTATIVE COURSE OUTLINE

| <i>LECTURE</i> | <i>DAY</i> | <i>DATE</i> | <i>TOPIC</i>                             | <i>READING</i> | <i>HOMEWORK DUE</i> |
|----------------|------------|-------------|------------------------------------------|----------------|---------------------|
| 1              | T          | 8/19        | INTRODUCTION TO CONTROL SYSTEMS          | CH. 1          |                     |
| 2              | R          | 8/21        | MODELING MECHANICAL SYSTEMS              | CH. 2.1        |                     |
| 3              | T          | 8/26        | LINEARIZATION AND SYSTEM REPRESENTATIONS | CH. 2.1-2.2    | #1                  |
| 4              | R          | 8/28        | MODELING ELECTROMECHANICAL SYSTEMS       | CH. 2.3        |                     |
| 5              | T          | 9/2         | MODELING CASE STUDY                      |                |                     |
| 6              | R          | 9/4         | LAPLACE TRANSFORM                        | CH. 3.1        | #2                  |
| 7              | T          | 9/9         | INVERSE LAPLACE TRANSFORMS               | CH. 3.1        |                     |
| 8              | R          | 9/11        | BLOCK DIAGRAMS                           | CH. 3.2        | #3                  |
| 9              | T          | 9/16        | SYSTEM RESPONSE                          | CH. 3.3        |                     |
| 10             | R          | 9/18        | SYSTEM RESPONSE                          | CH. 3.4        | #4                  |
| 11             | T          | 9/23        | SYSTEM RESPONSE                          | CH. 3.5-3.6    |                     |
|                | R          | 9/25        | <b>EXAM 1 (LECTURES 1-10)</b>            |                |                     |
| 12             | T          | 9/30        | INTRODUCTION TO CONTROL                  | CH. 4.1        |                     |
| 13             | R          | 10/2        | FEEDFORWARD AND FEEDBACK CONTROL         | CH. 4.2        | #5                  |
| 14             | T          | 10/7        | PID CONTROL                              | CH. 4.3        |                     |
| 15             | R          | 10/9        | PID CONTROL                              | CH. 4.3        | #6                  |
| 16             | T          | 10/14       | NO CLASS – FALL BREAK DAY                |                |                     |
| 17             | R          | 10/16       | PID CONTROL                              | CH. 4.3        | #7                  |
| 18             | T          | 10/21       | ROOT LOCUS                               | CH. 5.1-5.2    |                     |
| 19             | R          | 10/23       | ROOT LOCUS                               | CH. 5.1-5.2    | #8                  |
| 20             | T          | 10/28       | CONTROLLER DESIGN USING ROOT LOCUS       | CH. 5.3-5.4    |                     |
| 21             | R          | 10/30       | CONTROL DESIGN CASE STUDY                |                | #9                  |
| 22             | T          | 11/4        | LEAD AND LAG COMPENSATION                | CH 5.5         |                     |
|                | R          | 11/6        | <b>EXAM 2 (LECTURES 11-20)</b>           |                |                     |
| 23             | T          | 11/11       | FREQUENCY RESPONSE                       | CH. 6.1        |                     |
| 24             | R          | 11/13       | BODE DIAGRAMS                            | CH. 6.1        | #10                 |
| 25             | T          | 11/18       | BODE DIAGRAMS                            | CH. 6.1        |                     |
| 26             | R          | 11/20       | FREQUENCY RESPONSE DESIGN METHOD         | CH 6.4-6.7     |                     |
| 27             | T          | 11/25       | CONTROL DESIGN CASE STUDY                |                |                     |
|                | R          | 11/27       | NO CLASS- THANKSGIVING BREAK             |                |                     |
|                | T          | 12/2        | FINAL PRESENTATIONS                      |                |                     |
|                | R          | 12/4        | FINAL PRESENTATIONS                      |                |                     |